

Skill 2: OS Shell Implementation & Interactive Loop Testing

Lab Record & Terminal Execution Output

1. Task Description

To Create Main Loop, Display Prompt, Read User Input, Handle Exit Conditions, Design Control Flow Diagram, Test Interactive Loop.

2. Program 1: Basic Interactive Shell (shell.c)

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

int main() {
    char input[100];

    while (1) {
        // Display prompt
        printf("myshell> ");
        fflush(stdout);

        // Read user input
        if (fgets(input, sizeof(input), stdin) == NULL) {
            printf("
");
            break;
        }

        // Remove newline
        input[strcspn(input, "
")] = '\0';

        // Handle exit condition
        if (strcmp(input, "exit") == 0) {
            printf("Exiting shell...
");
            break;
        }

        // Handle empty input
        if (strlen(input) == 0) {
            continue;
        }

        // Display entered command
        printf("You entered: %s
", input);
    }
    return 0;
}
```

3. Terminal Session 1 (Basic Shell Execution)

```
rohit@LAPTOP-07A78KDI:~/OS$ nano shell.c
rohit@LAPTOP-07A78KDI:~/OS$ gcc shell.c -o shell
```

```

rohit@LAPTOP-07A78KDI:~/OS$ ./shell
myshell> hello
You entered: hello
myshell> linux
You entered: linux
myshell> test
You entered: test
myshell> exit
Exiting shell...
rohit@LAPTOP-07A78KDI:~/OS$ while (1) {
printf("myshell> ");
}
-bash: syntax error near unexpected token `{ '
-bash: syntax error near unexpected token "myshell> "
...: command not found
-bash: syntax error near unexpected token `}'
rohit@LAPTOP-07A78KDI:~/OS$ fgets(input, sizeof(input), stdin);
-bash: syntax error near unexpected token `input,'
rohit@LAPTOP-07A78KDI:~/OS$

```

4. Program 2: Raw Mode & Key Handling Shell (keyboard_shell.c)

```

#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <termios.h>
#include <string.h>

#define BUFFER_SIZE 100

void enable_raw_mode() {
    struct termios raw;
    tcgetattr(STDIN_FILENO, &raw);
    raw.c_lflag &= ~(ICANON | ECHO);
    tcsetattr(STDIN_FILENO, TCSAFLUSH, &raw);
}

void disable_raw_mode() {
    struct termios original;
    tcgetattr(STDIN_FILENO, &original);
    original.c_lflag |= (ICANON | ECHO);
    tcsetattr(STDIN_FILENO, TCSAFLUSH, &original);
}

int main() {
    char buffer[BUFFER_SIZE];
    int index = 0;
    char ch;

    enable_raw_mode();
    printf("myshell> ");
    fflush(stdout);

    while (1) {
        read(STDIN_FILENO, &ch, 1);

        // Handle Enter key
        if (ch == '\n' || ch == '\r') {
            buffer[index] = '\0';

```

```

        printf("
");

        if (strcmp(buffer, "exit") == 0) {
            printf("Exiting shell...
");
            break;
        }

        if (index > 0) {
            printf("Command entered: %s
", buffer);
        }
        index = 0;
        printf("myshell> ");
        fflush(stdout);
    }
    // Handle Backspace
    else if (ch == 127 || ch == ' ') {
        if (index > 0) {
            index--;
            buffer[index] = ' ';
            printf(" ");
            fflush(stdout);
        }
    }
    // Handle normal characters
    else if (ch >= 32 && ch <= 126) {
        if (index < BUFFER_SIZE - 1) {
            buffer[index] = ch;
            index++;
            printf("%c", ch);
            fflush(stdout);
        }
    }
}

disable_raw_mode();
return 0;
}

```

5. Terminal Session 2 (Keyboard Shell Execution)

```

rohit@LAPTOP-07A78KDI:~/OS$ nano keyboard_shell.c
rohit@LAPTOP-07A78KDI:~/OS$ gcc keyboard_shell.c -o keyboard_shell
rohit@LAPTOP-07A78KDI:~/OS$ ./keyboard_shell
myshell> hello
Command entered: hello
myshell> hello
Command entered: hello
myshell> hel
Command entered: hel
myshell> printf(" ");
Command entered: printf(" ");
myshell> helprintf(" ");
Command entered: helprintf(" ");
myshell> exit
Exiting shell...
rohit@LAPTOP-07A78KDI:~/OS$ char buffer [BUFFER_SIZE]
Command 'char' not found, did you mean:
  command 'chr' from snap chr (0.80.4)

```

```
command 'chat' from deb ppp (2.5.2-1+1.2)
command 'shar' from deb sharutils (1:4.15.2-13)
command 'car' from deb ucommon-utils (7.0.1-1)
command 'phar' from deb php8.5-cli (8.5.4-0ubuntu1.1)
command 'phar' from deb php-cli (2:8.5+99ubuntu1)
command 'chr' from deb chr (0.1.80-1)
command 'chr' from deb chr-tiny (0.1.80-1)
command 'pchar' from deb pchar (1.5-7)
See 'snap info <snapname>' for additional versions.
rohit@LAPTOP-07A78KDI:~/OS$
```